

Treatment Algorithms for Anogenital Warts (AGW)

It is recognized that there are multiple treatment algorithms for anogenital warts (AGW), with selection influenced by various factors. For instance, if clinic staff are experienced in ablative techniques, the protocol may involve using ablation as the first-line treatment for all patients regardless of wart burden, followed by immunotherapy. In the UK, this approach is common, as it minimizes the number of required clinic visits.

Management of Large AGW

Large anogenital warts—defined as lesions too extensive for local trichloroacetic acid (TCA) or cryotherapy—should be referred to a specialist. Based on clinical experience, initial pre-treatment with immunotherapy (e.g., imiquimod 5%) for up to 16 weeks is recommended to stimulate an immune response and reduce recurrence risk. Evidence supports this strategy: patients pre-treated with imiquimod before surgical excision have lower long-term recurrence rates compared to those undergoing surgery alone. Data on other immunotherapies in this context remain limited.

Following immunotherapy, large AGW should be surgically excised under general anesthesia. If residual or recurrent warts develop, immunotherapy should be reinitiated. Histopathological examination of the excised tissue is essential to rule out verrucous carcinoma or squamous cell carcinoma.

Guidance for Daily Practice

Practical guidance for managing common clinical scenarios in AGW, along with recommended actions, is summarized in clinical practice tables.

Preventing AGW

Anogenital warts can be effectively prevented with HPV vaccination. The **quadrivalent vaccine** (targeting HPV types 6, 11, 16, and 18) and the **nonavalent vaccine** (targeting HPV 6, 11, 16, 18, 31, 33, 45, 52, and 58) protect against the HPV types responsible for most AGW, cervical cancer, and other anogenital and oropharyngeal cancers.

The quadrivalent vaccine has demonstrated up to **100% efficacy** in preventing vaccine-type HPV-associated AGW in women. Following its introduction in Australia, a 4-year follow-up study reported a **59% reduction** in AGW prevalence among young females. A concomitant, though less pronounced, decline in AGW incidence was observed in heterosexual men, likely due to herd protection.

HPV vaccination has the potential to significantly reduce healthcare costs and alleviate the workload in sexual health clinics. The vaccine is effective in boys aged 12–15 years and is licensed for use in both males and females in most countries where it is available.

While the role of HPV vaccination in the **treatment** of existing AGW remains unclear, emerging data suggest potential benefit even in individuals previously exposed to HPV.

Conclusions

The guidance outlined supports clinicians in the diagnosis and management of AGW in routine practice, aiming to improve patient health and quality of life. The proposed treatment approach is flexible, allowing for adaptation based on local resources, clinician expertise, and patient preferences.